

Dancing with your Thumbs Market and Usability Analysis

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Having finally finished *Dancing with your Thumbs*, I'm very happy with the overall design. The goal of this project was to create a game that was fun, engaging, and polished, and I believe I succeeded on all three goals. Fun and engagement are hard to quantify, but while testing the game on the DE1-SoC, I often found myself playing for much longer than I originally intended, and certainly longer than was necessary to ensure the features were all there. I consider that a success unto itself.

Polish, however, is a bit more objective. It can be broken up into two components – does everything behave as expected, and is that behavior satisfying? This project underwent multiple revisions throughout development to ensure that the final result felt fully polished. At first, there was only one row of lights, but that felt strange to see, so I added an additional two rows of lights for the “good” tiles, this time green to differentiate them from the yellow “perfect” lights. It was hard to tell which row of lights you hit, so I added a momentary flash upon scoring points with a color based on the score (yellow for +2, green for +1). The score goes up to 9999, as I imagined that would be by far large enough for any player, but just in case it caps at 9999 rather than overflowing to 0000. All these behaviors have been thoroughly tested, and the machine behaves well.

Another factor to consider is ergonomics. A common source of frustration in early testing was that on higher difficulties, the machine would often spawn a light in just after a player scored. This resulted in a lot of accidental button presses, as the user had not yet removed their finger from the button. I fixed this by forcing lights to wait for the flash to end before spawning in, which helped quite a bit.

The final consideration I made was size. Ultimately, this design was a lot larger than any other in 271. Early designs had the size over 800 – and that was before a few of the bonus features were added! I managed to bring the overall size down to 524, which I'm happy with. There may still be some places where I could optimize size a bit, but ultimately for how much this design does, this size seems reasonable.